

Ga'u echo: A Digital Shrine for Remote Haptic Emotional Connection

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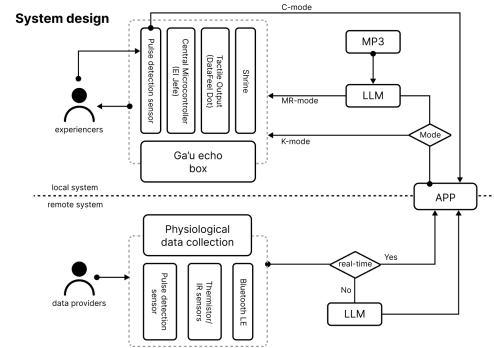
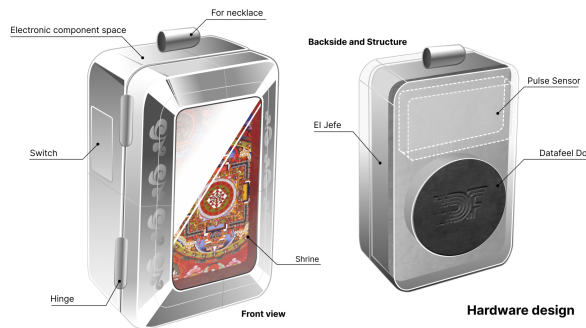


Figure 1: The Hardware and system concept design of Ga'u echo

Abstract

Remote emotional connection and internal emotional regulation play a critical role in addressing emotional isolation and mental stress in modern society. This study presents “Ga’u Echo,” a wearable device grounded in biosensing and affective computing technologies, specifically developed to facilitate remote emotional connection and support individual emotional regulation. Functioning as a “digital ga’u box” for the user, “Ga’u Echo” transforms the physiological rhythms of distant loved ones and the user’s own internal states into tangible tactile feedback. This enables individuals to form deeper, more authentic emotional connections with both others and themselves.

CCS Concepts

• Human-centered computing → Mobile devices; Haptic devices.

Keywords

Emotional connection, Emotional reification, Traditional culture, Tactile interaction, Remote interaction

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1 Introduction

Social relationships exert a profound influence on physical health, mental well-being, and overall life satisfaction[11]. Research by Foster et al. has demonstrated that even monthly social interactions with family and friends can significantly reduce mortality risk[4]. Furthermore, contemporary lifestyles are increasingly characterized by digitalization and social isolation, which intensifies individuals’ longing for meaningful emotional connections and spiritual well-being[5, 6]. Physical touch serves as a crucial medium for emotional bonding[9]; however, current communication technologies fall short in effectively encoding affective tactile information, thereby limiting the potential for meaningful remote tactile interaction[3].

In addressing universal needs such as emotional comfort and interpersonal connection, the “ga’u box” in Tibetan culture offers a profound and non-technological historical precedent. The ga’u box is a portable, miniature shrine that typically contains sacred objects such as a small statue of the Buddha, religious texts, personal items of deceased loved ones, or consecrated relics (e.g., blessed pills or relics from revered lamas). It functions as a talisman, offering spiritual protection and emotional strength to the wearer[2, 10]. As an ‘affective interface’, the ga’u box materializes abstract emotional sentiments into tangible, portable forms[1].

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Advancements in affective computing and wearable technology have opened new possibilities for the evolution of such emotional interfaces. Affective computing focuses on enabling computational systems to detect, interpret, respond to, and simulate human emotions[7], while wearable devices employ biosensors such as those measuring heart rate, electrodermal activity, and skin temperature to continuously capture multimodal physiological signals associated with emotional states. These signals are then translated into real-time, adaptive feedback through tactile, auditory, or visual modalities[8].

This project represents a digital extension of that emotional interface. Our Ga'u Echo integrates biosensing capabilities with data-driven tactile feedback technology to transform traditional spiritual comfort into a platform for real-time emotional exchange with distant loved ones and for personalized emotional regulation. It aims to explore a novel interaction paradigm that redefines individualized spiritual and emotional connections in the digital era.

2 System Design

Ga'u Echo is a personal wearable that utilizes multimodal haptic technology to enhance remote emotional connection and perceived presence among distant relatives by simulating heartbeats. It serves as both an embodiment of remote users' vital signs and a personal health/spiritual companion for local users, enabling novel sensory interaction beyond audiovisual modalities. Its asymmetric design distinguishes data providers from experiencers, assigning distinct roles and capabilities. This supports flexible, authentic emotional connections adaptable to diverse lifestyles and contexts. This section details system components for both user types and briefly outlines the supported embodied interactions.

2.1 Remote System

The primary objective of the remote system is to seamlessly and accurately capture physiological data from remote users (data providers) and convert it into a digital signal suitable for transmission. The system primarily consists of a wearable sensing unit and a backend AI model.

2.1.1 Physiological Data Collection: Remote users wear lightweight sensors (e.g., wristband/skin patch) capturing.

- Heart Rate: Real-time monitoring via PPG sensor (optical capillary pulsation detection).
- Body Temperature: Continuous tracking using thermistor/IR sensors.
- Data Transmission: Secure transmission via Bluetooth LE to the mobile app.

2.1.2 AI-Driven Personalized Physiological Database: To ensure continuity in emotional connection, the system incorporates a generative AI model.

- Personalized Database: Builds user-specific vector DB from longitudinal physio-data (HR/Temp) using RNNs/ Transformers, encoding unique rhythmic patterns across emotional states.
- Data Generation: Generates a "digital heartbeat" (AI-simulated physio-stream) during offline periods to maintain emotional continuity.

2.1.3 Software Ecosystem: Mobile app serves as core platform for cross-user system integration, data routing and UI control (mode selection/personalization).

2.2 Local System

Core emotional experience platform converting digital signals to multisensory feedback. Hardware reinterprets traditional "ga'u box" with embedded electronics.

2.2.1 Gau Box Enclosure: Culturally-inspired design preserving Tibetan aesthetics. Integrates sacred space for mementos (photos/relics/amulets) and Housing for electronic components.

2.2.2 Central Microcontroller (El Jefe): The El Jefe microcontroller is responsible for receiving data from the mobile application via Bluetooth, decoding the incoming physiological signals, and orchestrating the tactile output of the DataFeel Dot module.

2.2.3 Tactile Output (DataFeel Dot): The DataFeel Dot module generates precise, nuanced, and programmable vibration waveforms that accurately replicate the rhythm, intensity, and intermittent variations of a heartbeat, creating a highly realistic tactile sensation. Additionally, based on the received temperature data, the device modulates its surface temperature to simulate the warmth of a real heartbeat, thereby enhancing the tactile experience with emotional warmth and intimacy.

2.2.4 Local User Biometric Sensing: To sense the user's physiological state, the device integrates a photoplethysmography (PPG) sensor for continuous heart rate variability (HRV) monitoring. This enables the system to infer emotional stress and adapt its behavior in response to the user's real-time state.

3 Workflow

3.1 Kinship Mode (K-mode):

Simulates remote family members' presence by converting their physio-data (e.g., heartbeat) into thermal/vibration patterns. Uses recorded data when offline. Leverages LLM to generate personalized frequency bands for preserving unique cardiac signatures.

3.2 Calm Mode (C-mode):

Triggers DataFeel via mobile app when detecting local user's emotional spikes. Delivers user-defined thermal-tactile cues (e.g., cooling/vibration) to facilitate emotional regulation and breathing focus.

3.3 Meditation and Recitation Mode (MR-mode):

Transforms audio frequency bands (meditation/recitation) into synchronized thermal-rhythmic feedback. Enhances mindfulness by inducing meditative states through multisensory stimulation.

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